Abirami Kumar

(510) 309-9098 | abirami.kumar@berkeley.edu | linkedin.com/in/-abiramikumar | abirami07.github.io

EDUCATION

University of California, Berkeley **Berkeley, CA**
B.A. Data Science & B.A. Computer Science *Aug 2025 – May 2028*

- **Relevant Coursework:** Data Structures, Principles and Techniques of Data Science, Linear Algebra & Differential Equations, Discrete Mathematics, Computational Structures in Data Science
- **Student Organizations:** Nova Consulting, Software Product, Data Discovery Program, CS Kickstart

EXPERIENCE

Software Engineering Intern **Remote**
AP0110 *May 2026 – Aug 2026*

- Built a natural-language extraction pipeline for a React-based AI assistant, parsing user inputs, normalizing units, validating physiological ranges, and detecting symptom negation to support reliable user interactions.
- Engineered a JavaScript rules engine evaluating 6 physiological metrics, implementing deterministic risk classification, missing-data handling, and symptom-aware validation across edge cases.
- Developed a Python/Pandas ETL pipeline integrating 3 NHANES datasets into 4,907 validated adult records, implementing data cleaning, risk classification, and structured data exports.

AI Researcher **Berkeley, CA**
KnaiTai Open Initiative *August 2026 – Present*

- Developed Python-based knowledge activation and runtime-selection logic within the A4 architecture, enabling context-aware evidence retrieval and dynamic AI reasoning.
- Contributed to automated regression and integration testing using **Pytest** across a suite of 57 passing tests, validating runtime operations, evidence freshness, and authorization decisions.
- Debugged stale evidence propagation in multi-cycle reasoning workflows, implementing state synchronization fixes and validating system behavior through end-to-end execution traces.

Technical Analyst **Remote**
Cisco | Nova Consulting *September 2026 – Present*

- Developing a structured competitive benchmarking framework to evaluate channel-partner profitability across enterprise technology vendors, analyzing margin structures, rebates, incentives, and service revenue models.
- Analyzing vendor documentation, financial disclosures, and partner-program data to identify competitive trends and translate findings into actionable recommendations for enterprise stakeholders.

PROJECTS

Lexia | UC Berkeley AI Hackathon | GitHub **June 2026**

- Built and demoed a full-stack AI accessibility platform in 48 hours using **React, FastAPI, and Python**, dynamically adapting web content based on real-time reading behavior.
- Engineered AI workflows integrating Browserbase, Stagehand, Claude, and Deepgram for browser automation, content extraction, LLM-based comprehension, and voice interaction.
- Implemented gaze-based adaptation using PyGaze, analyzing fixation duration, regressions, and line-skipping signals to trigger personalized content transformations.

Sather Sense: Indoor Navigation System | 1st Place at SodaHacks | GitHub **March 2026**

- Built a cross-platform indoor navigation application using React Native, TypeScript, and Expo, delivering real-time voice guidance and haptic navigation for visually impaired users.
- Engineered a React/Vite administration interface for floor-plan ingestion, spatial annotation, and navigation graph management.
- Implemented graph-based pathfinding across multi-floor building layouts and OCR-assisted localization for indoor positioning.

SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL
Frontend: React, React Native, Expo, Vite
Frameworks & Tools: FastAPI, REST APIs, Git, Linux, Pytest, Pandas, NumPy